

Fetal Health Classification from Cardiotocogram Data

1. Problem Statement

A cardiotocogram (CTG) records the fetal heart rate and uterine contractions during pregnancy. Reading it by hand is slow and subjective. The task is to classify each CTG record as **Normal**, **Suspect** or **Pathological** from its numeric features. We reimplemented the reference paper and checked whether our accuracies matched the ones it reports.

2. Dataset

UCI Cardiotocography data (*fetal_health.csv*): 2,126 records and 21 features, each labelled by expert obstetricians. After preprocessing, 2,113 records were used and split 70/30 into 1,479 training and 634 test samples. The classes are imbalanced (about 78% Normal):

	Normal	Suspect	Pathological	Total
Train	1,149	202	128	1,479
Test	497	90	47	634

Features were reduced to 10 through feature selection and standardised with StandardScaler.

3. Approach

We trained ten classifiers on the same train/test split, using library defaults where the paper does not give hyperparameters. Each model is scored on test accuracy, with per-class precision, recall and F1 and confusion matrices as supporting views. Accuracies are then compared with the values reported in the paper.

4. Implementation

Python in Google Colab with scikit-learn, XGBoost and LightGBM. Preprocessing and feature selection write processed train and test CSV files, which the two model notebooks read. Notebook 03 trains an MLP (one hidden layer of 100 units), Gradient Boosting, XGBoost, LightGBM and a Linear SVM. Notebook 04 trains Logistic Regression, KNN, Decision Tree, Random Forest and an RBF SVM. Both notebooks end with a comparison against the paper.

5. Results (test accuracy, %)

Model	Ours	Paper	Difference (pp)
LightGBM	95.90	95.58	+0.32
Gradient Boosting	95.58	95.58	+0.00
XGBoost	95.58	94.95	+0.63
Random Forest	95.27	94.95	+0.32
MLP	94.95	93.53	+1.42
KNN	93.85	94.01	-0.16
Decision Tree	93.22	93.38	-0.16
SVM (RBF)	91.96	92.11	-0.15
Logistic Regression	89.59	89.59	+0.00
Linear SVM	89.27	89.27	+0.00

6. Conclusions

- Boosted trees and Random Forest did best: LightGBM reached 95.90% (608 of 634 correct), followed by Gradient Boosting and XGBoost at 95.58%. The linear models stayed near 89%.
- Nine of the ten models are within 0.65 points of the paper. The MLP differs most (+1.42), probably because the paper does not state its network size or random seed. With 634 test samples, one sample is 0.16 points, so the small gaps are within noise.
- Suspect is the hardest class. Even LightGBM recalls only 78.9% of Suspect cases, against 99.0% for Normal, because the class is small and overlaps with Normal.
- Limitation: all results come from a single split without cross-validation.